



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE VENKATESHWARA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

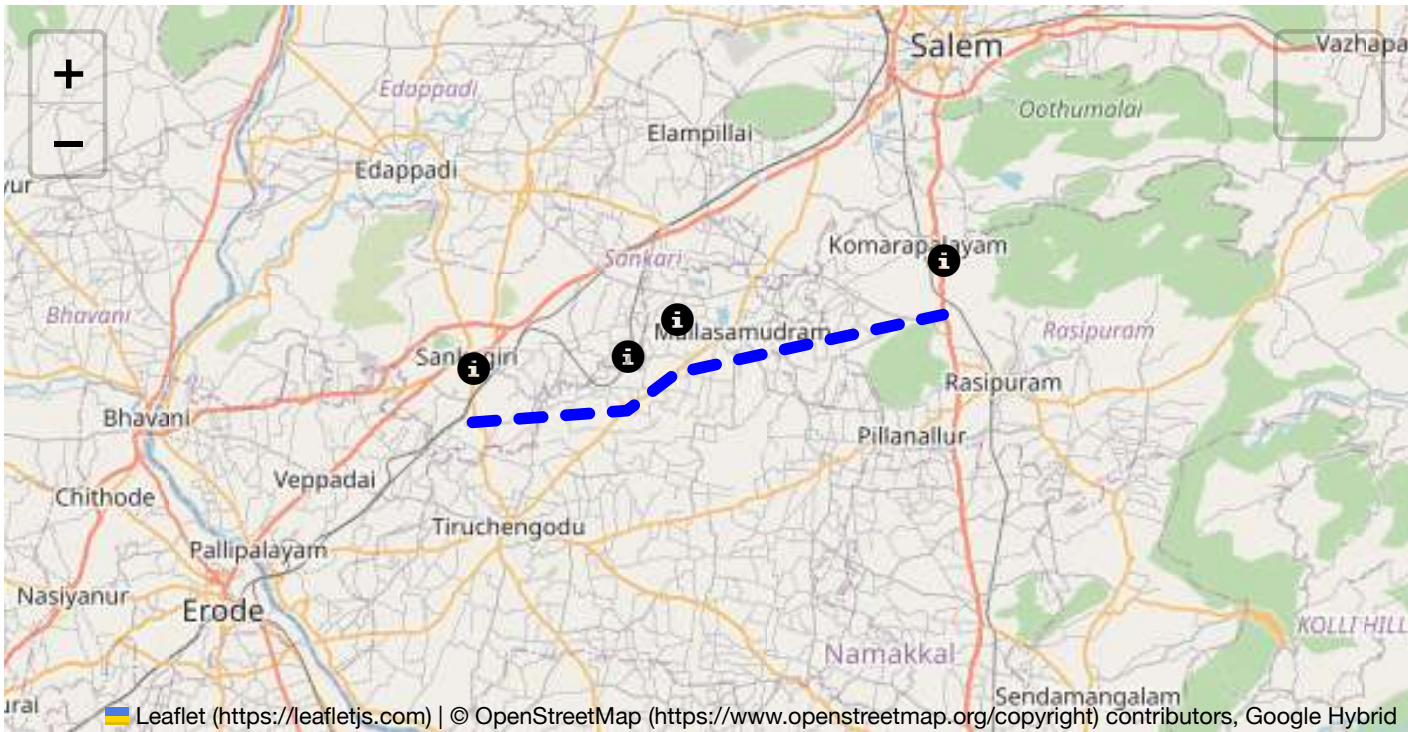
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

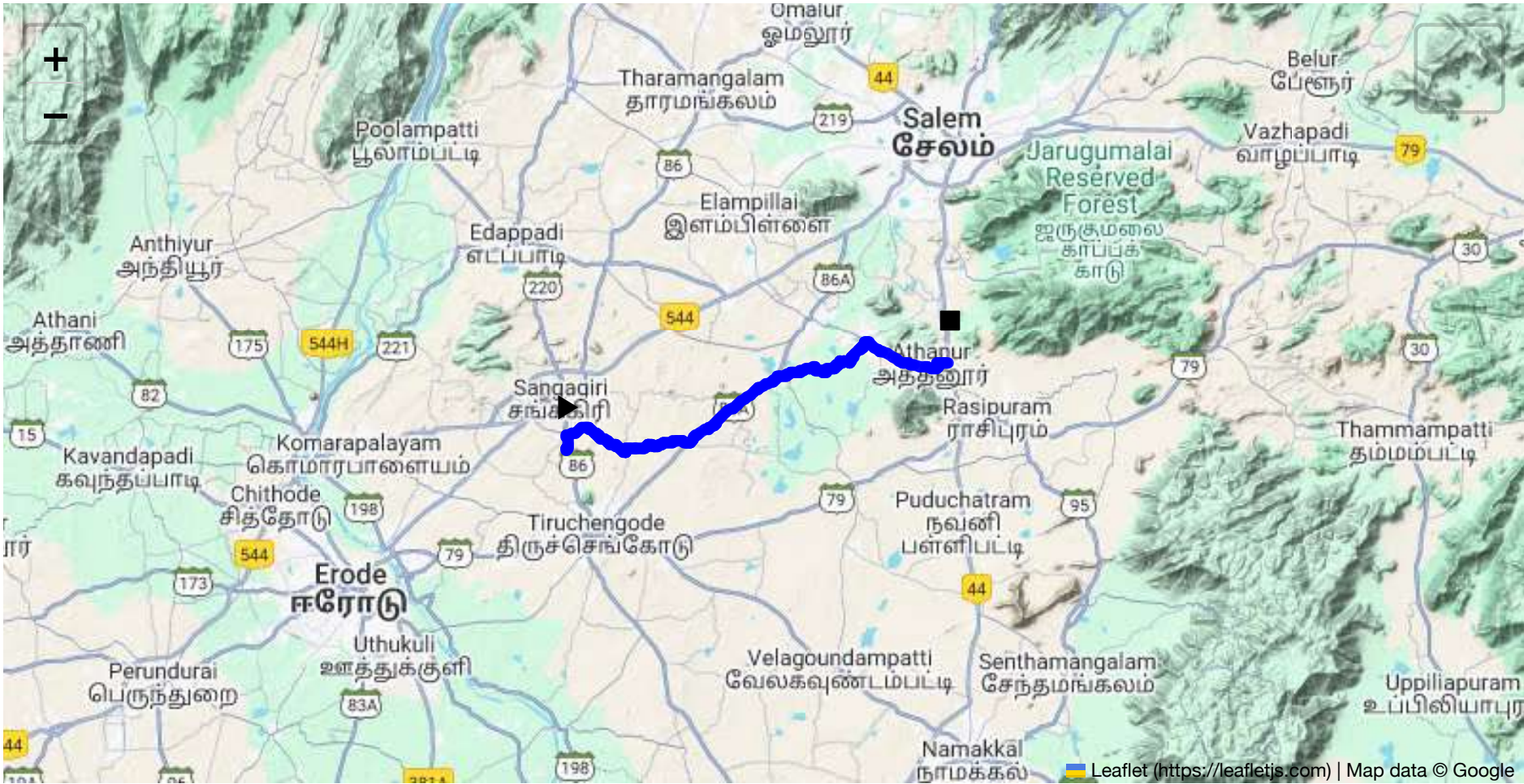
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 37.96 km
Estimated Duration: 1.0 hours

Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.49964, 78.149565)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report: Sangagiri to Rasipuram via Morepalayam and Sembampalyam

- Overview of the Route Map:** The route from Sangagiri to Rasipuram covers approximately 37.96 kilometers, passing through intermediate points at Morepalayam and Sembampalyam. The path primarily follows rural roads, with a mix of state highways and local roads. This provides connectivity through agricultural and semi-urban areas.
- Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu typically experiences a tropical climate with a peak in rainfall during the northeast monsoon (October to December). Potential weather hazards include waterlogging and reduced visibility during heavy rains, particularly in low-lying areas. Windstorms can also occur, affecting driving conditions.
- Analysis of Traffic Patterns:** Traffic congestion can occur in morning and evening peak hours, particularly near urban centers and market areas. Sangagiri and Rasipuram typically experience higher traffic volumes during these times. The rural segments may face slower-moving agricultural vehicles, impacting traffic flow.
- Assessment of Road Quality and Infrastructure:** The state highways are generally well-maintained, although rural roads may have uneven surfaces, potholes, and narrower lanes. Signage might be insufficient, and lighting can be sporadic, especially outside urban areas, increasing nighttime travel risks.
- Suggestions for Alternative Routes in Emergencies:** For alternative routes, the SH-79 and SH-95 offer parallel and intermediary paths that can be used to bypass main congestion points or in case of road blockages. These highways have better facilities and accessibility for emergency detours.
- Summary of Local Regulations Affecting Hazardous Material Transport:** Tamil Nadu mandates that all vehicles transporting hazardous materials adhere to specific guidelines, including mandatory

display of warning signage, route permits, and compliance with weight limits. Transport during peak hours might be restricted in urban areas to mitigate risks.

- 7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** There have been occasional incidents involving heavy vehicles due to driver fatigue or mechanical failures. While not specific to this route, similar rural routes have experienced issues related to sharp turns and narrow bridges.
- 8. **Environmental Considerations and Sensitive Areas:** The route traverses areas with active agricultural activities; care should be taken to prevent contamination from spillages. Proximity to water bodies in the area necessitates awareness of the environmental impact of hazardous spills.
- 9. **Analysis of Communication Coverage:** The communication infrastructure is generally reliable near urban centers. However, certain rural segments may experience weak mobile network coverage, particularly in low-density areas, impacting real-time communication and navigation.
- 10. **Estimated Emergency Response Times for Different Route Segments:** Response times are typically longer in rural areas due to the distance from emergency services based in urban nodes. Main towns like Sangagiri and Rasipuram have faster response capabilities, generally within 20-30 minutes, while rural areas may have delays extending to 45 minutes or more.
- 11. **Overall Summary of Risk Assessment:** This route presents moderate risk levels due to traffic patterns, rural road conditions, and variable weather. Proper planning and adherence to regulations can mitigate many risks associated with transporting hazardous materials. Emergency preparedness, including route familiarization and alternative plans, is crucial for minimizing potential operational disruptions.

In conclusion, while there are challenges associated with uneven roads, traffic congestion, and climate-related issues, a well-prepared truck driver can navigate this route safely with the appropriate precautions and awareness of local conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.45478, 77.88925	30 KM/Hr
5	Turn	High	11.43673, 77.91703	15 KM/Hr
6	Turn	High	11.43926, 77.91705	15 KM/Hr
7	Turn	Medium	11.43930, 77.92282	30 KM/Hr
8	Turn	Medium	11.43967, 77.92345	30 KM/Hr
9	Turn	Medium	11.43979, 77.93009	30 KM/Hr
10	Turn	Medium	11.44030, 77.93066	30 KM/Hr
11	Turn	Medium	11.44067, 77.93203	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.44088, 77.93223	15 KM/Hr
13	Turn	Medium	11.44127, 77.93218	30 KM/Hr
14	Turn	Medium	11.44163, 77.93244	30 KM/Hr
15	Turn	High	11.44243, 77.93244	15 KM/Hr
16	Turn	Medium	11.44285, 77.93440	30 KM/Hr
17	Turn	Medium	11.44093, 77.93821	30 KM/Hr
18	Turn	Medium	11.44550, 77.95329	30 KM/Hr
19	Turn	Medium	11.44369, 77.96036	30 KM/Hr
20	Turn	Medium	11.44391, 77.96277	30 KM/Hr
21	Turn	High	11.49063, 78.02680	15 KM/Hr
22	Turn	High	11.49135, 78.03252	15 KM/Hr
23	Turn	High	11.49183, 78.03257	15 KM/Hr
24	Turn	Medium	11.49355, 78.03703	30 KM/Hr
25	Turn	Medium	11.49410, 78.03864	30 KM/Hr
26	Turn	Medium	11.49381, 78.04241	30 KM/Hr
27	Turn	Medium	11.49637, 78.04986	30 KM/Hr
28	Turn	High	11.49676, 78.05010	15 KM/Hr
29	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
30	Turn	High	11.49553, 78.05338	15 KM/Hr
31	Turn	Medium	11.49547, 78.05442	30 KM/Hr
32	Turn	High	11.49518, 78.05498	15 KM/Hr
33	Turn	Medium	11.49544, 78.05538	30 KM/Hr
34	Turn	Medium	11.49526, 78.05677	30 KM/Hr
35	Blind Spot	Blind Spot	11.49548, 78.05904	10 KM/Hr
36	Turn	High	11.49469, 78.05908	15 KM/Hr
37	Turn	High	11.49451, 78.05995	15 KM/Hr
38	Turn	High	11.49314, 78.05979	15 KM/Hr
39	Turn	High	11.49305, 78.05987	15 KM/Hr
40	Turn	Medium	11.49306, 78.06011	30 KM/Hr
41	Turn	Medium	11.49358, 78.06081	30 KM/Hr
42	Turn	High	11.49370, 78.06142	15 KM/Hr
43	Turn	High	11.49561, 78.06192	15 KM/Hr
44	Turn	High	11.49665, 78.06789	15 KM/Hr
45	Turn	High	11.49725, 78.06787	15 KM/Hr
46	Turn	Medium	11.49738, 78.06806	30 KM/Hr
47	Turn	High	11.49738, 78.06877	15 KM/Hr
48	Turn	High	11.49906, 78.06934	15 KM/Hr
49	Turn	Medium	11.49969, 78.07116	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
50	Turn	High	11.50124, 78.07191	15 KM/Hr
51	Turn	Medium	11.50070, 78.07343	30 KM/Hr
52	Turn	Medium	11.50092, 78.07374	30 KM/Hr
53	Turn	High	11.50082, 78.07865	15 KM/Hr
54	Turn	Medium	11.50143, 78.07938	30 KM/Hr
55	Turn	High	11.50374, 78.08048	15 KM/Hr
56	Turn	High	11.50424, 78.08040	15 KM/Hr
57	Turn	Medium	11.51303, 78.08909	30 KM/Hr
58	Blind Spot	Blind Spot	11.51452, 78.08915	10 KM/Hr
59	Turn	High	11.51417, 78.09091	15 KM/Hr
60	Turn	Medium	11.51445, 78.09107	30 KM/Hr
61	Turn	Medium	11.51468, 78.09148	30 KM/Hr
62	Blind Spot	Blind Spot	11.51494, 78.09165	10 KM/Hr
63	Turn	Medium	11.51485, 78.09173	30 KM/Hr
64	Blind Spot	Blind Spot	11.49693, 78.14237	10 KM/Hr
65	Turn	High	11.49711, 78.14191	15 KM/Hr
66	Blind Spot	Blind Spot	11.49991, 78.14224	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45478, 77.88925



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43673, 77.91703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43930, 77.92282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43967, 77.92345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44067, 77.93203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44163, 77.93244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44093, 77.93821



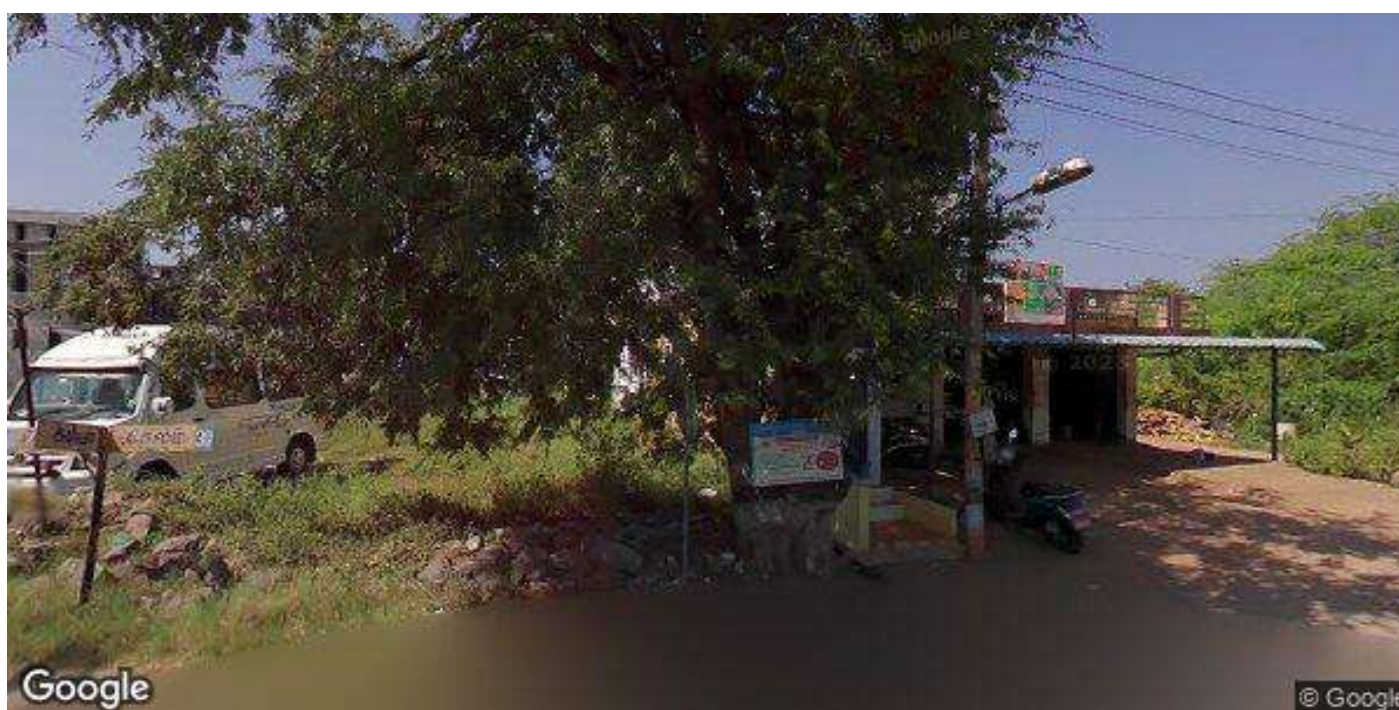
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44369, 77.96036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44391, 77.96277



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49063, 78.02680



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49135, 78.03252



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49183, 78.03257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49355, 78.03703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49410, 78.03864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49381, 78.04241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49637, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49676, 78.05010



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49553, 78.05338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49547, 78.05442



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49518, 78.05498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49544, 78.05538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49526, 78.05677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49548, 78.05904



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 78.05908



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 78.05995



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49314, 78.05979



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49305, 78.05987



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49306, 78.06011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49358, 78.06081



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49370, 78.06142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49561, 78.06192



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49665, 78.06789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49725, 78.06787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49738, 78.06806



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49738, 78.06877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49906, 78.06934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49969, 78.07116



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50124, 78.07191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50070, 78.07343



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50092, 78.07374



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50082, 78.07865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50143, 78.07938



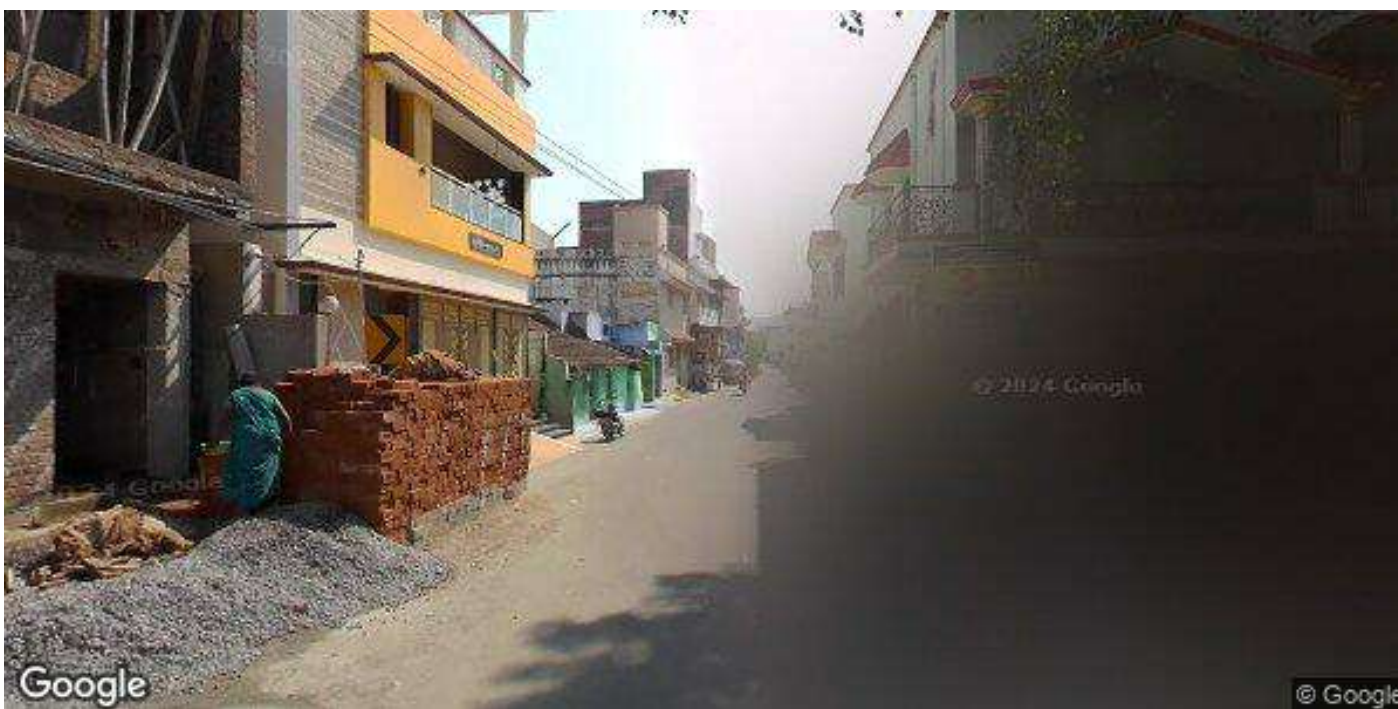
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50374, 78.08048



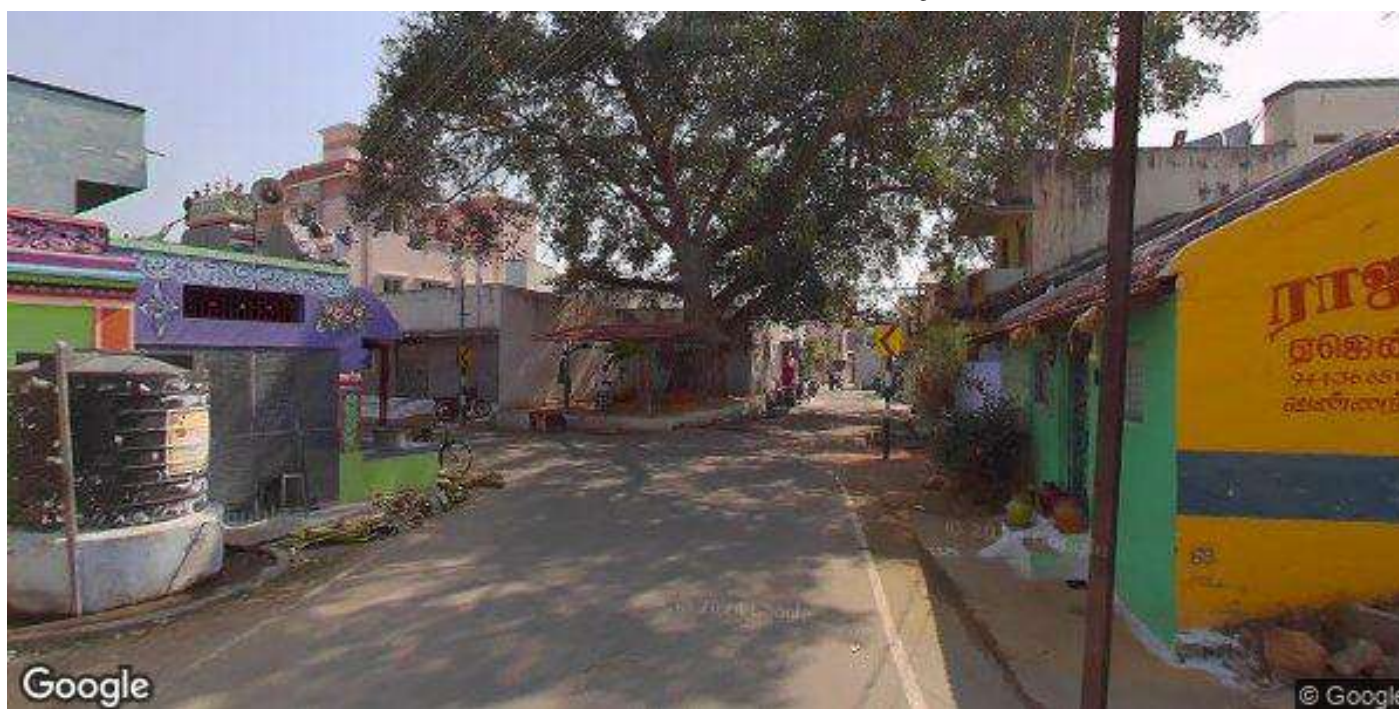
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50424, 78.08040



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51303, 78.08909



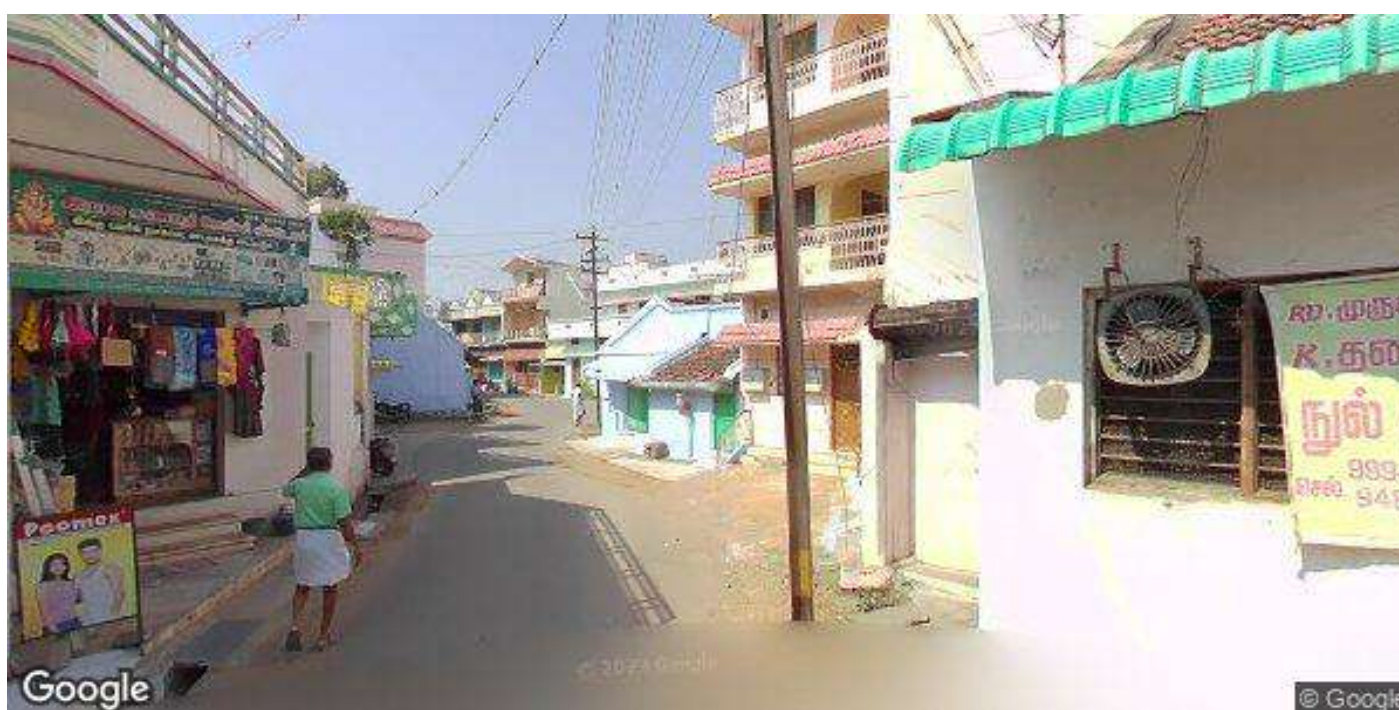
Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51452, 78.08915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51417, 78.09091



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51445, 78.09107



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51468, 78.09148



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51494, 78.09165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51485, 78.09173



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49693, 78.14237



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49711, 78.14191



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49991, 78.14224

